



Operating Instructions

Gas engine-generator set – Power container
Series 4000

SS180139/01E



A Rolls-Royce
solution

Product name	Model number	Engine model	Application group
MTU 16V4000 GS	GG16V4000A1	16V4000L64	3A, continuous operation, unrestricted
MTU 16V4000 GS	GG16V4000A1	16V4000L32 16V4000L32FB 16V4000L33	3A, continuous operation, unrestricted
MTU 16V4000 GS	GG16V4000A2	16V4000L32 16V4000L32FB 16V4000L33	3A, continuous operation, unrestricted
MTU 20V4000 GS	GG20V4000A1	20V4000L32 20V4000L32FB 20V4000L33	3A, continuous operation, unrestricted
MTU 20V4000 GS	GG20V4000A2	20V4000L32 20V4000L32FB 20V4000L33	3A, continuous operation, unrestricted

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Table of Contents

1	Preface		4.2	Engine-Generator Set in Container	45
1.1	Preface	5	4.2.1	Application group	45
2	Safety		4.2.2	Nameplate and designation	46
2.1	Important provisions for all products	6	4.2.3	Container	51
2.2	Gas systems - Intended use	8	4.2.4	Engine-generator set	55
2.3	Personnel and organizational requirements	10	4.2.5	Engine	56
2.4	Safety regulations for initial start-up and operation	11	4.2.6	Generator and coupling	57
2.5	Safety regulations for commissioning and operation, specific information for gaseous fuel supply or and dual fuel applications	13	4.3	Monitoring and Control	58
2.6	Safety regulations for assembly, maintenance, and repair work	15	4.3.1	MMC and MIP controllers	58
2.7	Safety regulations for assembly, maintenance and servicing work, specific information for gaseous fuel and dual fuel supply or applications	19	4.4	Charge-air System and Gas System	60
2.8	Fire and environmental protection, indirect materials	20	4.4.1	Charge-air system and gas system - Engine-generator set	60
2.9	Fire prevention and environmental protection, fluids and lubricants, auxiliary materials, specific information for gaseous fuel supply in gaseous-fuel or dual-fuel applications	23	4.4.2	Gas system - Container	61
2.10	Standards for warning notices in the text and highlighted information	24	4.5	Exhaust System	63
2.11	Conventions for warning notices on safety signs on the engine-generator set	25	4.5.1	Exhaust system - Engine-generator set	63
2.12	Safety signs on engine-generator set - Overview	27	4.5.2	Exhaust silencer	64
2.13	Safety plates on container - Overview	29	4.6	Lube Oil System	65
2.14	Overview of protective devices on the engine-generator set	32	4.6.1	Lube oil system - Engine-generator set	65
3	Transport		4.6.2	Lube oil system - Container	66
3.1	Container standards	34	4.7	Cooling System and Heating Water System	69
3.2	Transport	35	4.7.1	Cooling system - Engine-generator set	69
3.3	Transport locking device for engine mounting	37	4.7.2	Cooling system - Container	70
3.4	Transport locking device - Generator mounting	38	4.7.3	Coolant cooler	72
3.5	Crankshaft transport locking device	40	4.8	Ventilation System	75
3.6	Tie-down instructions	42	4.8.1	Ventilation system	75
4	Functional Description		4.9	Electrical System	76
4.1	Chapter overview	44	4.9.1	Generator circuit breaker (GCB) panel	76
			4.9.2	Starter battery	77
			4.9.3	Battery charger	78
			5	Technical Data	
			5.1	Technical data	79
			6	Operation	
			6.1	Container - Operation	80
			7	Troubleshooting	
			7.1	Fault messages	81
			8	Maintenance	
			8.1	Maintenance activities - Engine-generator set	82
			8.2	Maintenance activities - Container	83

9 Task Description		9.5.3 Insulation - Check	98
9.1 Chapter overview	84	9.6 Cooling System and Heating Water System	99
9.2 Container	85	9.6.1 Cooler - Check	99
9.2.1 Container - External contamination check	85	9.6.2 Fan - Cleaning	100
9.2.2 Container - Door seal check	86	9.6.3 Cooling system - Leak check	101
9.2.3 Container - Door hinge check	87	9.6.4 Coolant pressure - Check	102
9.2.4 Ladders and handrails - Check	88	9.6.5 Engine coolant - Sample extraction and analysis	103
9.2.5 Safety plates - Check	89	9.7 Electrical System	104
9.3 Ventilation	90	9.7.1 Grounding point - Check	104
9.3.1 Weather guard - External contamination check	90	9.7.2 Heating, connector sockets and lighting - Function check	105
9.3.2 Fan - Check	91		
9.3.3 Fan - Cleaning	92	10 Appendix A	
9.3.4 Louver dampers - Function check	93	10.1 Abbreviations	106
9.4 Gas System	94	10.2 Conversion tables	108
9.4.1 Gas system - Gas lines leak check	94	10.3 Contact person/Service partner	112
9.4.2 Gas alarm system - Function check	95		
9.5 Exhaust System	96	11 Appendix B	
9.5.1 Exhaust silencer - Condensate drain check	96	11.1 Index	113
9.5.2 Flue bracing - Check	97		

1 Preface

1.1 Preface

These Operating Instructions contain general instructions for the proper operation of your engine-generator set from the manufacturer MTU. It is vitally important that all persons working on or with the engine-generator set are familiar with all the contents of this manual and follow the instructions provided carefully at all times.

In some cases, it may prove necessary to depart from the guidelines specified in this manual in response to extraneous circumstances. Any such departure must comply with locally applicable directives and standards. All relevant safety directives and safety precautions shall be duly considered and respected.

Following these Operating Instructions will ensure that the plant operates efficiently and reliably.

All procedural instructions and safety precautions shall be followed closely to ensure that the plant operates as intended and to avoid injury or harm. Read and follow the instructions in the "Safety" section at the beginning of this manual.

All the instructions and diagrams have been checked to ensure accuracy and user-friendliness, however the skills and qualification of the operating personnel remain the most significant factor nevertheless. MTU cannot give any assurances for the actual results of any work performed according to the descriptions provided in this manual, nor shall MTU be held liable for any resultant personal injury or material damage. Personnel charged with running and operating the plant do so entirely at their own risk.

Other documents included in the scope of supply shall also be given due consideration in addition to these Operating Instructions.

All documents required by Machinery Directive 2006/42/EC are included in the standard scope of documentation for the engine-generator set. The following documents are supplied in addition:

- Test records
- Spare parts catalog
- Wiring diagrams

Definition of MTU

MTU refers to Rolls-Royce Power Systems AG and MTU Friedrichshafen GmbH or an affiliated company pursuant to Section § 15 AktG (German Stock Corporation Act) or a controlled company (joint venture).

2 Safety

2.1 Important provisions for all products

General

This product may pose a risk of injury or damage in the following cases:

- Incorrect use
- Operation, maintenance and repair by unqualified personnel
- Changes or modifications which are neither made nor authorized by the manufacturer
- Noncompliance with the safety instructions and warning notices

Nameplates

The product is identified by nameplate, model designation or serial number. This data must match the specifications in these instructions.

Nameplates, model designation or serial number can be found on the product.

All EU-certified engines delivered by MTU come with a second nameplate. This second nameplate is delivered "loosely" with the engine. If the nameplate secured to the engine after installation in the vehicle/system is not visible without the removal of components, the system integrator must install the second nameplate in a clearly visible area on the vehicle/system.

Emissions regulations and emissions label

Responsibility for compliance with emissions regulations

Modification or removal of any mechanical/electronic components or the installation of additional components including the execution of calibration processes that might affect the emissions characteristics of the product are prohibited by emissions regulations. Emissions-related components must only be serviced, exchanged or repaired if the components used for this purpose are approved by the manufacturer.

Noncompliance with these specifications will invalidate the design type approval or certification issued by the emissions regulation authorities. The manufacturer does not accept any liability for violations of the emissions regulations.

The product must be operated over its entire life cycle according to the conditions defined as "Intended use" (→ Page 8).

Emissions certification applicable to engines with EPA Nonroad Tier 4 emissions certification in accordance with 40 CFR 1039 and with EPA Locomotive emissions certification in accordance with 40 CFR 1033

Extract from the standard:

Failing to follow these instructions when installing a certified engine in a piece of nonroad equipment violates federal law (40 CFR 1068.105(b)), subject to fines or other penalties as described in the Clean Air Act.

Extract from the standard:

If you install the engine in a way that makes the engine's emission control information label hard to read during normal engine maintenance, you must place a duplicate label on the equipment, as described in 40 CFR 1068.105.

When fitting the second label, the requirements of 40 CFR 1068.105(c) must be followed and observed. This paragraph describes the process for requesting and fitting the label, the documentation obligations and storage obligations for the required documents.

Replacing components with emissions labels

On all engines from MTU fitted with emissions labels, these labels must remain on the engine throughout its operational life.

Exception: Engines used exclusively in land-based, military applications other than by US government agencies.

Please note the following when replacing components with emissions labels:

- The relevant emissions labels must be affixed to the spare part.
- Emissions labels shall not be transferred from the replaced part to the spare part.
- The emissions labels must be removed from the replaced part and destroyed.

2.2 Gas systems – Intended use

Intended use

The machine is intended to be used exclusively for electrical power generation and waste heat utilization in stationary operation, either in sufficiently ventilated rooms or in container solutions.

A gas/air mixture is fed to the internal combustion reciprocating piston engine. This is converted into mechanical work in the individual cylinders through combustion. The flywheel of the engine drives the synchronous generator connected via an elastic coupling, resulting in a conversion to electric energy. The engine and the generator are joined to the base frame with resilient, vibration-damped elements. For mains frequency 60 Hz, depending on the series either a gearbox is used between the engine and generator (Series 4000) or the engine speed is increased accordingly (Series 400). The waste heat generated during the combustion process is dissipated to the surroundings via coolers (power generator) and/or made available to the customer via heat exchangers (combined heat and power plant).

Area of application	Applicable	Comment
Industry	Yes	–
Business/Service	Yes	In some individual cases, however, such as with swimming pool facilities, schools, heating stations for apartment buildings and the like, no end user in the classic sense is involved
Household	No	Operation is excluded
Potentially explosive area	No	Operation is not permitted in potentially explosive areas
Underground	No	Operation is excluded
Mobile use	No	Operation is excluded

Table 1: Area of application of machine

There are 3 operating modes of the machine:

- Mains parallel operation
- Mains failure operation
- Isolated operation

Foreseeable misuse

Any other use, particularly misuse, is considered as being contrary to the intended purpose. Such improper use increases the risk of injury and damage when working with the product. The manufacturer shall not be held liable for any damage resulting from improper use.

Operation

Operation is not permissible if the physical conditions specified on the technical data sheet or on/in other documents (concerning temperature, pressure, site altitude, etc.) are not complied with. Operation is not permissible without protective equipment, usage of approved spare parts, exclusion of conversions, warning of foreseeable misuse.

Operation is only permissible with the use of fluids and lubricants (natural gas, biogas, quality-compliant gas, quality-compliant water, lubricants) in accordance with the (→ Fluids and Lubricants Specifications).

Operation is only permissible with preservation agents approved by MTU in accordance with the (→ Preservation and Represervation Specifications).

The specifications of the manufacturer will be amended or supplemented as necessary. Prior to operation, make sure that the latest version of these are adopted. The latest version can be found at:

- <http://www.mtu-solutions.com>

Surroundings, climate, temperature

The weather plays no role with regard to the intended use of the engine-generator set/module provided that operation takes place inside an installation room. For container applications, however, the permissible outdoor temperatures at the installation site play a decisive role. The product-specific ambient conditions such as climate and temperature are indicated in the technical description.

Prerequisite for proper use

Fulfillment of all the requirements set out in the "Installation conditions" document.

Any use that does not fall within the described limits for use is regarded as improper.

Work which requires communication with the control units must only be carried out with a diagnosis software or with tools approved by the manufacturer.

Modifications or conversions

Unauthorized changes to the product breach the conditions concerning its intended use and compromise safety.

Only modifications or conversions expressly authorized by the manufacturer are regarded as compliant in this sense. The manufacturer shall not be held liable for any damage resulting from unauthorized modifications or conversions.

2.3 Personnel and organizational requirements

Organizational measures of the user/manufacturer

This manual must be issued to all personnel involved in operation, maintenance, repair, assembly, installation, or transportation.

Keep this manual handy in the vicinity of the product such that it is accessible to operating, maintenance, repair, assembly, installation, and transport personnel at all times.

Personnel must receive instruction on product handling and repair based on this manual. In particular, personnel must have read and understood the safety requirements and warnings before starting work.

This is important in the case of personnel who only occasionally perform work on or around the product. Such personnel must be instructed repeatedly.

Personnel requirements

All work on the product must be carried out by trained, instructed and qualified personnel only:

- Training at the Training Center of the manufacturer
- Qualified personnel from the areas mechanical engineering, plant construction, and electrical engineering and also for work with live parts

The operator must define the responsibilities of the personnel involved in operation, maintenance, repair, assembly, installation, and transport in writing.

Personnel shall not report for duty under the influence of alcohol, drugs or strong medication.

Clothing and personal protective equipment

Always wear appropriate personal protective equipment, e.g. safety shoes, ear protectors, protective gloves, goggles, breathing mask. Follow the instructions concerning personal protective equipment in the respective descriptions of the individual activities or the manufacturer's documentation included in the delivery.

When working on live components, in particular on electrical switchgear or on high-voltage component, wear special protective clothing according to IEC 61482.

Safe handling of Substances of Very High Concern pursuant to the REACH regulation (Registration, Evaluation, Authorization and restriction of CHemicals): We recommend wearing protective gloves at all times in order to reduce risk when working.